Schema inference and entity identification

Status and result. This is a partial submission: five dataset-audit verdicts remain unresolved. All 50 link verdicts are human-verified by the user. Discovery retrieved **638** catalogue records and code-ranked **50** datasets. Six agent-generated mappings produce **77,860 field observations**, **110 proposed links**, **107 profile-eligible links** and **50 company profiles**. The source appendix lists all 50 catalogue URLs; README provides run commands.

Discovery and acquisition. Deterministic metadata scoring rewards explicit identifiers, suppliers, procurement and registers, and penalizes aggregate/geographic content. Scores are uncalibrated ranking signals. The national catalogue returned HTTP 403; Queensland and NSW supplied the pool. Downloadability, shape and overlap guided selection from the shortlist. The processed set spans five publishers but is concentrated in Queensland and procurement, limiting representativeness. Official DataStore representations bypassed challenged download routes; manifests preserve original/acquired URLs, licence, timestamps and hashes.

Source	Acquired representation	Records	Observations
QBCC contractors	Nested JSON	30,000	53,428
Transport and Main Roads contracts	CSV	1,000	885
System Procurement contracts	CSV	202	151
Suitable Operator Register	Nested JSON	15,111	23,029
P&SCO contracts	CSV	44	88
Treasury Corporation contracts	CSV	279	279

Runtime agent. Eight stages inspect bytes, choose a parser, profile training rows, propose a mapping, execute withheld rows, critique/revise, revalidate and present approval evidence. Proposal and critique are separate model calls. Errors trigger bounded repair; fingerprints/checkpoints retain completed work across failures. A seventh source supplies raw bytes and acquisition metadata, not a mapping. One engine executes all configs; unsupported shapes stop for adapter review. The fixed 0.8 field-confidence floor cannot be lowered by the agent. Human review sees raw examples, rationales, errors, unmapped fields, semantic risks and licences, and approves exact mapping hashes.

Evidence and hard source. Ingestion preserves source claims independently, with traceable snapshot/row identities, licence, extractor/config identity and distinct observation/ingestion times. Metadata publication-time proxies are labelled. Source, field and link confidence remain separate. The operator register mixes ACN-prefixed and other values: label-aware checksum diagnostics let the agent repair extraction. Semantics remain the weakest step; checksums cannot establish subject meaning. The initial 30% holdout becomes debugging evidence after revision. A reusable adapter is justified after repeated generic-reader failures, rather than per-source scripts.

Cost and demo. Actual Haiku spend is **$8.0687**, checked by the user; per-source dollar allocation is unavailable. Receipts record 543,341 input and 93,554 output tokens across 46 responses, a lower bound after early failed attempts. Final QBCC onboarding used eight stages, two calls, 27,946 input / 3,597 output tokens and 26.61 seconds. Per-record model API cost is $0; compute billing is unknown. The historical 59/60-request ledger is preserved. Demo has a separate, non-resettable six-request/$1 allowance within the remaining $10 cap. A synthetic end-to-end test proves proposal, revision and review-packet completion; a live seventh-source run still requires credentials.

Identity policy. Versioned pairwise links cite raw-record endpoints, canonical ABN/ACN key, rule score and evidence. Threshold **0.98** admits checksum-valid exact ABN (score 0.995) or ACN (0.985), with corroborating normalized names and compatible subject/type/capacity. Rule scores are not precision estimates. Only explicit corporate own-capacity ABN/ACN co-occurrence bridges namespaces; suffix inference and name-only links abstain. Employees, trustees, conflicting identifiers and unsafe compound parties are refused. Incompatible supported subjects cannot be unioned into one profile. Rejected endpoints and dependent bridges remain quarantined. Parent contracts supply no ownership evidence, so ownership edges remain absent.

Link evaluation. The submitted matcher has **50 explicit verdicts, 50 correct, precision 100% at threshold 0.98**. The sample reuses 44 previously checked raw-record pairs and adds six seeded current proposals. The user confirmed human verification of all 50 verdicts. Final attribution is 50 human; earlier AI-prepared assessments and per-item reasoning are retained as provenance. This is a targeted development sample, not independent population precision; it does not establish that every proposed link is correct. `outputs/audit_summary_submission.json` records the denominator, threshold and reviewer mix. Approval is recorded separately from correctness.

Dataset evaluation remains incomplete. Of the frozen 20-item shortlist sample, 12 are positive, three negative and **five inconclusive**. Conditional false-positive rate is **3/15 = 20%**, with 75% verification coverage. The full 20-item rate is unknown, bounded by 15–40%; no 20-item point estimate is claimed. Unavailable/empty source responses and missing PDF/API records remain evidence gaps, not negative labels.

Profiles and withdrawal. Each field preserves confidence, winning provenance, alternatives and selection reasons. Reconciliation ranks source reliability, distinct-source support, mapping confidence and publication proxy, with stable tie-breaks; duplicate rows are not independent corroboration. Conflicting field confidence is capped at 0.7. Current selected profiles contain no conflicting fields; conflict retention is tested. Removing QBCC changes 50 profiles and loses 99 fields; removing the operator register changes 46 and loses 46. These are fixed-membership counterfactuals: production withdrawal must also retract links and rebuild graph membership.

Engineering and scale. Offline replay reconstructs hash-checked snapshots. Regression tests cover parser/JSONL limits, checksums, provenance, approval tampering, repair, budget isolation, truncated-response cache rejection and quarantine. Billable usage is saved before rejecting malformed/truncated responses. At 500 sources, review capacity, provider throttling and non-transactional local job/budget state become bottlenecks. Use identifier indexes, idempotent queues, bounded retries, transactional state and partitioned storage.

Migration and three-week priorities. Retain snapshots, ontology/config/engine hashes, model/prompt receipts, thresholds, evidence and labels. Shadow-replay affected blocks, compare splits/merges and independent labelled precision, approve migration, retract superseded edges and rebuild affected profiles with rollback. Preserve canonical-key aliases rather than silently replacing identity. Priorities are resolving the five dataset verdicts, running a credentialed unseen-source demo, calibrating confidence and implementing graph-aware source withdrawal. Detailed validation and rendering evidence accompany this two-page report.